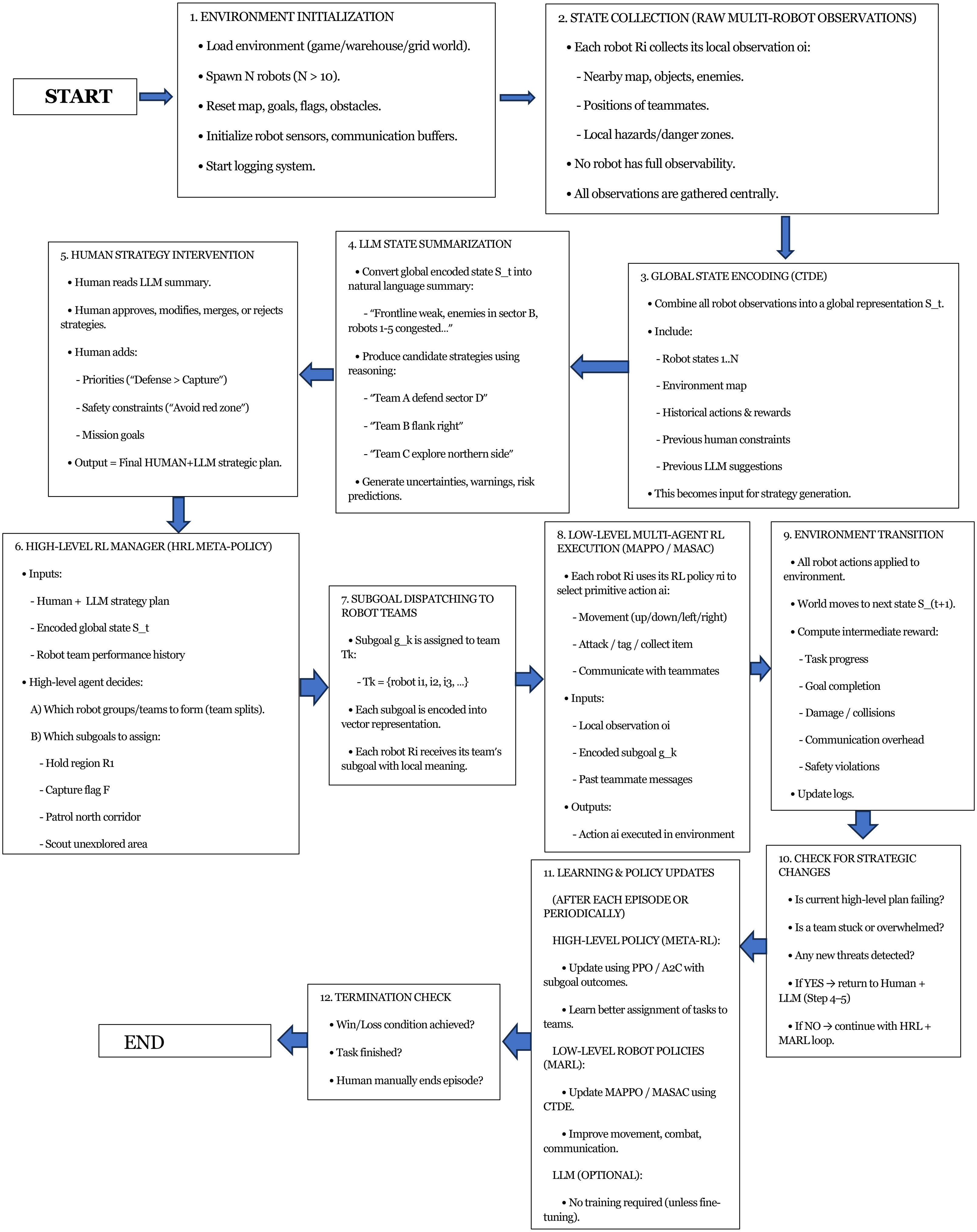

START
1. ENVIRONMENT INITIALIZATION
• Load environment (game/warehouse/grid world).
• Spawn N robots (N > 10).
• Reset map, goals, flags, obstacles.
• Initialize robot sensors, communication buffers.
• Start logging system.
2. STATE COLLECTION (RAW MULTI-ROBOT OBSERVATIONS)
• Each robot Ri collects its local observation oi:
- Nearby map, objects, enemies.
- Positions of teammates.
- Local hazards/danger zones.
• No robot has full observability.
• All observations are gathered centrally.
3. GLOBAL STATE ENCODING (CTDE)
• Combine all robot observations into a global representation S_t.
• Include:
- Robot states 1..N
- Environment map
- Historical actions & rewards
- Previous human constraints
- Previous LLM suggestions
• This becomes input for strategy generation.
4. LLM STATE SUMMARIZATION
• Convert global encoded state S_t into natural language summary:
- "Frontline weak, enemies in sector B, robots 1-5 congested..."
• Produce candidate strategies using reasoning:
- "Team A defend sector D"
- "Team B flank right"
- "Team C explore northern side"
• Generate uncertainties, warnings, risk predictions.
5. HUMAN STRATEGY INTERVENTION
• Human reads LLM summary.
• Human approves, modifies, merges, or rejects strategies.
• Human adds:
- Priorities ("Defense > Capture")
- Safety constraints ("Avoid red zone")
- Mission goals
• Output = Final HUMAN+LLM strategic plan.
6. HIGH-LEVEL RL MANAGER (HRL META-POLICY)
• Inputs:
- Human + LLM strategy plan
- Encoded global state S_t
- Robot team performance history
• High-level agent decides:
A) Which robot groups/teams to form (team splits).
B) Which subgoals to assign:
- Hold region R1
- Capture flag F
- Patrol north corridor
- Scout unexplored area
7. SUBGOAL DISPATCHING TO ROBOT TEAMS
• Subgoal g_k is assigned to team Tk:
- Tk = {robot i1, i2, i3, ...}
• Each subgoal is encoded into vector representation.
• Each robot Ri receives its team's subgoal with local meaning.
8. LOW-LEVEL MULTI-AGENT RL EXECUTION (MAPPO / MASAC)
• Each robot Ri uses its RL policy πi to select primitive action ai:
- Movement (up/down/left/right)
- Attack / tag / collect item
- Communicate with teammates
• Inputs:
- Local observation oi
- Encoded subgoal g_k
- Past teammate messages
• Outputs:
- Action ai executed in environment
9. ENVIRONMENT TRANSITION
• All robot actions applied to environment.
• World moves to next state S_(t+1).
• Compute intermediate reward:
- Task progress
- Goal completion
- Damage / collisions
- Communication overhead
- Safety violations
• Update logs.
10. CHECK FOR STRATEGIC CHANGES
• Is current high-level plan failing?
• Is a team stuck or overwhelmed?
• Any new threats detected?
• If YES → return to Human + LLM (Step 4–5)
• If NO → continue with HRL + MARL loop.
11. LEARNING & POLICY UPDATES
(AFTER EACH EPISODE OR PERIODICALLY)
HIGH-LEVEL POLICY (META-RL):
• Update using PPO / A2C with subgoal outcomes.
• Learn better assignment of tasks to teams.
LOW-LEVEL ROBOT POLICIES (MARL):
• Update MAPPO / MASAC using CTDE.
• Improve movement, combat, communication.
LLM (OPTIONAL):
• No training required (unless fine-tuning).
12. TERMINATION CHECK
• Win/Loss condition achieved?
• Task finished?
• Human manually ends episode?
END